

BRIDGING THE GAP: ADDITIVE STRUCTURE IN THE TWIN PRIME COFACTORS

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ABSTRACT. We present a conjecture, which we call the bridging conjecture (BC): for every $V \in \mathbb{N}$ there exist $u, v, w \in A002822$ with $u \leq v \leq V < w$ and $u + v = w$, where $A002822$ denotes the set of positive integers w such that $6w - 1$ and $6w + 1$ are both prime (twin prime witnesses). We establish its relationships to the Twin Prime Conjecture and Dubner’s conjectures — BC is equivalent to the Twin Prime Conjecture assuming Dubner’s middle number conjecture, and follows unconditionally from Dubner’s sumset conjecture — and to other conjectures about gaps between primes, twin primes, and their witnesses: BC implies a gap bound $w_{n+1} - w_n \leq w_n$, placing it in the same family as Oppermann’s conjecture ($g(p) < \sqrt{p}$ for ordinary primes) and Selcoe’s interval conjecture (an $O(w_n)$ gap bound by a different mechanism), and consequently $\pi_2(x) \gg \log x$. We also re-express Dubner’s exception set A243956 in set-theoretic terms as $D = \mathbb{N}_{\geq 1} \setminus \{i + j : i, j \in A002822\}$, which explains the additive origin of its elements rather than merely listing them, and reformulates Dubner’s sumset conjecture as the assertion that D is finite.

1. INTRODUCTION

A *twin prime pair* is a pair of primes $(p, p + 2)$. The integer $p + 1$ between them is divisible by 6; write $p + 1 = 6w$, so w is a positive integer with $6w - 1$ and $6w + 1$ both prime. The set of such witnesses,

$$A002822 = \{w \in \mathbb{N} : 6w - 1 \text{ and } 6w + 1 \text{ are both prime}\},$$

is OEIS sequence A002822 [9]. The Twin Prime Conjecture asserts that $A002822$ is infinite. A prime belonging to a twin prime pair is called a *t-prime*.

In 1996, Stephen Wagler communicated a question to Paulo Ribenboim: what is the smallest middle number that is not the sum of two t-primes? Ribenboim asked Harvey Dubner to investigate [3, pp. 291–299]. Dubner’s paper [2] is the primary published source; he formulated four conjectures in response, which we recall in Section 2. Wagler’s

question is closest in spirit to Conjecture 2.2, which asks whether every middle number greater than 6 is the sum of two t-primes. In Section 3 we re-express Dubner's exception set A243956 in set-theoretic terms as the complement of the sumset of A002822 in $\mathbb{N}_{\geq 1}$, which explains the additive origin of the exceptions rather than merely listing them, and reformulates Conjecture 2.3 as the assertion that this complement is finite. In Section 4 we introduce the bridging conjecture and establish its relationships to the Twin Prime Conjecture and Dubner's conjectures.

2. DUBNER'S CONJECTURES

Dubner [2] stated four conjectures, which we recall here with descriptive names.

Conjecture 2.1 (Dubner: middle number conjecture). Every middle number greater than 6 is the sum of two middle numbers.

In terms of A002822: a middle number $6w$ is the sum of two middle numbers $6u$ and $6v$ if and only if $w = u + v$ with $u, v \in A002822$. So Conjecture 2.1 is equivalent to: every $w \in A002822$ with $w > 1$ is the sum of two smaller elements of A002822. Dubner verified this up to 2×10^{10} .

Conjecture 2.2 (Dubner: t-prime sumset conjecture). Every middle number greater than 6 is the sum of two t-primes.

Conjecture 2.3 (Dubner: sumset conjecture). Every multiple of 6 greater than 4206 is the sum of two middle numbers.

Conjecture 2.4 (Dubner: t-prime pair sumset conjecture). Every even number greater than 4208 is the sum of two t-primes.

Conjecture 2.3 is strictly stronger than Conjecture 2.1: it requires all sufficiently large multiples of 6 to decompose, not only those that are middle numbers. The sumset reformulation of Conjecture 2.3 is developed in Section 3.

3. THE SUMSET OF A002822 AND DUBNER'S EXCEPTION SET

The natural object here is the sumset of A002822 with itself. For any $i, j \in A002822$ the sum $i + j$ is a positive integer, and this sum need not itself belong to A002822: it may lie in A002822 (a middle number decomposition of $i + j$) or in A067611 (the complement). In either case $i + j$ is an element of $\mathbb{N}_{\geq 1}$ produced by the pair (i, j) . Iterating over all pairs in A002822 and collecting their sums therefore generates a subset of $\mathbb{N}_{\geq 1}$ with no restriction on whether the sums are twin prime

witnesses. Let D denote the complement of this sumset in $\mathbb{N}_{\geq 1}$, i.e. the set of positive integers admitting no such representation; this is OEIS A243956 [10]. Dubner identified eleven multiples of 6 that are not the sum of two middle numbers,

$$\{96, 402, 516, 786, 906, 1116, 1146, 1266, 1356, 3246, 4206\},$$

and conjectured these are the only exceptions. A243956 is derived from Dubner's set by dividing each element by 6 to recover the witness and adjoining 1 (which lies in A002822 but cannot be expressed as a sum of two elements of A002822), giving

$$D = \{1, 16, 67, 86, 131, 151, 186, 191, 211, 226, 541, 701\}$$

up to 2×10^{10} , and Conjecture 2.3 states that D is exactly this finite set. Conjecture 2.3 is phrased in terms of multiples of 6 and the threshold 4206 because Dubner identified the exceptions by exhaustive search and could only describe them by listing them. The sumset of A002822 provides an intrinsic characterisation: an integer belongs to D if and only if it is not reachable from A002822 by addition, with no prior knowledge of D required.

Definition 3.1. $D = \mathbb{N}_{\geq 1} \setminus \{i + j : i, j \in A002822\}$, the complement of the sumset of A002822 in $\mathbb{N}_{\geq 1}$. Since $\min(A002822) = 1$, every element of the sumset is at least 2, so $1 \in D$ automatically despite $1 \in A002822$.

By definition, D and $\{i + j : i, j \in A002822\}$ partition $\mathbb{N}_{\geq 1}$. The known value of D up to 2×10^{10} is A243956 [10].

Conjecture 3.2 (Conjecture 2.3, set-theoretic form). The complement of $\{i + j : i, j \in A002822\}$ in $\mathbb{N}_{\geq 1}$ is finite; equivalently, D is finite.

Let T denote the set of t-primes (OEIS A007534 [11]), i.e. primes belonging to a twin prime pair. Every sum of two t-primes is even. Analogously, Conjecture 2.4 has the following sumset reformulation.

Conjecture 3.3 (Conjecture 2.4, set-theoretic form). The complement of $\{p + q : p, q \in T\}$ in the positive even integers is finite.

Remark 3.4. Dubner's computation confirms that every element of D greater than 1 lies in A067611: none of the eleven exceptions beyond 1 is a twin prime witness. The element 1 lies in A002822 (since 5 and 7 are prime) but belongs to D because no two elements of A002822 sum to 1. Once A002822 is given, D is determined by its additive structure alone, without further reference to primality. Note also that Definition 3.1 is robust to the discovery of new exceptions: a new element of D beyond 701 would falsify Conjecture 2.3 (which fixes D as the known finite

set) but would merely enlarge D in Definition 3.1, leaving the partition intact.

4. THE BRIDGING CONJECTURE

Conjecture 4.1 (bridging conjecture (BC)). For every $V \in \mathbb{N}$ there exist $u, v, w \in A002822$ with $u \leq v \leq V < w$ and $u + v = w$.

Throughout this paper, references to “the bridging conjecture” mean Conjecture 4.1. That is, one can always find $u, v \in A002822$ with $u, v \leq V$ whose sum $w = u + v$ also lies in $A002822$ and exceeds V .

Theorem 4.2. *Conjecture 2.3 implies Conjecture 4.1, assuming the Twin Prime Conjecture.*

Proof. Given V , let w be the smallest element of $A002822$ greater than V (which exists by the Twin Prime Conjecture). For large enough V , $6w > 4206$, so Conjecture 2.3 gives $w = u + v$ with $u, v \in A002822$. By minimality of w , both u and v are at most V , hence $u, v \leq V$. $\square$

Theorem 4.3. *Conjecture 2.1 and the Twin Prime Conjecture together imply Conjecture 4.1.*

Proof. Given V , let w be the smallest element of $A002822$ greater than V . By Conjecture 2.1, $w = u + v$ for some $u, v \in A002822$ with $u, v < w$. By minimality of w , we have $u, v \leq V$. $\square$

Theorem 4.4. *Conjecture 4.1 implies the Twin Prime Conjecture.*

Proof. For any V , the bridging conjecture gives $w \in A002822$ with $w > V$. Since V is arbitrary, $A002822$ is infinite. $\square$

Corollary 4.5. *Assuming Conjecture 2.1,*

$$\text{Conjecture 4.1} \iff \text{Twin Prime Conjecture.}$$

Remark 4.6. The role of Conjecture 2.1 in Corollary 4.5 is asymmetric. The forward direction (bridging conjecture $\Rightarrow$ Twin Prime Conjecture) holds unconditionally: for any V , the bridging conjecture produces $w \in A002822$ with $w > V$, so $A002822$ is unbounded (Theorem 4.4). The reverse direction requires Conjecture 2.1: let w be the smallest element of $A002822$ above V ; Conjecture 2.1 decomposes $w = u + v$ with $u, v \in A002822$, and minimality of w gives $u, v \leq V$. Without Conjecture 2.1, the Twin Prime Conjecture alone does not guarantee that the next element above V is reachable as a sum of two smaller elements of $A002822$, and the reverse direction fails.

Theorem 4.7 (Witness Gap Bound). *Assuming Conjecture 4.1, consecutive elements $w_n < w_{n+1}$ of A002822 satisfy*

$$w_{n+1} - w_n \leq w_n.$$

Proof. Set $V = w_n$. The bridging conjecture gives $u, v \in A002822$ with $u \leq v \leq w_n$ and $w = u + v \in A002822$ with $w > w_n$, so $w_{n+1} \leq w = u + v \leq 2w_n$, hence $w_{n+1} - w_n \leq w_n$. Equality holds when $u = v = w_n$. $\square$

Remark 4.8. Conjecture 2.1 is not required for Theorem 4.7: the bound $w \leq 2w_n$ is an immediate consequence of the bridging conjecture's statement alone. What Conjecture 2.1 adds is qualitative: it asserts that *every* element of A002822 beyond 1 is reachable as a sum of two smaller elements of A002822, not only the specific w produced by the bridging conjecture.

Corollary 4.9. *Assuming Conjecture 4.1, the counting function $\pi_2(x) = |\{w \in A002822 : w \leq x\}|$ satisfies $\pi_2(x) \gg \log x$.*

Proof. By Theorem 4.7, every interval $(w_n, 2w_n]$ contains an element of A002822. Applying this inductively from $w_1 = 1$ gives at least $\lfloor \log_2 x \rfloor$ elements up to x . $\square$

5. DISCUSSION

Reformulation of Dubner's conjectures. Dubner's original statements of Conjectures 2.3 and 2.4 are phrased in terms of explicit numerical thresholds (4206 and 4208) derived from exhaustive computation. We reformulate them in Conjectures 3.2 and 3.3 as assertions that the complements of the relevant sumsets are finite. This rephrasing has two advantages. First, it removes the dependence on specific threshold values, which are artefacts of how far the computation was carried rather than features of the underlying mathematics. Second, it places Conjectures 2.3 and 2.4 in the same conceptual frame as Definition 3.1: all three are statements about the complement of a sumset in $\mathbb{N}_{\geq 1}$, making the structural parallel between them explicit.

The bridging conjecture as a splitting strategy. The bridging conjecture offers a way to split the problem. Rather than proving Conjecture 2.3 directly — which requires all sufficiently large multiples of 6 to decompose as sums of two middle numbers — one may instead attempt to establish the two weaker statements independently: Conjecture 2.1 (middle numbers themselves decompose) and the bridging conjecture (A002822 is unbounded, witnessed by sums of pairs of its own elements). Each of these makes a more focused demand on the

additive structure of $A002822$, and either might be more tractable individually than Conjecture 2.3 in full generality.

It is worth noting that Conjecture 2.1 alone does not imply the Twin Prime Conjecture: $A002822$ could be finite with every element (beyond the smallest) expressible as a sum of two smaller elements, and Conjecture 2.1 would still hold vacuously beyond that point. The bridging conjecture supplies exactly what is missing: it asserts that for every V , some pair of elements of $A002822$ below V sums to an element of $A002822$ above V , forcing $A002822$ to be unbounded. In this sense the bridging conjecture can be read as a bootstrap principle: if $A002822$ is known to be non-empty up to V , it certifies the existence of a further element beyond V .

Relationship to D and Conjecture 2.3. The bridging conjecture is also strictly weaker than Conjecture 2.3 in a second respect. Conjecture 2.3 makes a precise global claim: D is exactly the finite set A243956. The bridging conjecture makes no claim about the size or composition of D — it requires only that the sumset $\{i + j : i, j \in A002822\}$ is unbounded, which is independent of whether D is finite. It does, however, place one indirect constraint: since the bridging conjecture requires infinitely many elements of $A002822$ to appear as sums of pairs of smaller elements, D cannot eventually absorb all positive integers.

Conditional lower bound on twin prime density. Corollary 4.9 gives a conditional lower bound on the twin prime counting function. The gap bound $w_{n+1} - w_n \leq w_n$ implies $\pi_2(x) \gg \log x$ assuming BC. For comparison, the best unconditional upper bound is Brun’s theorem [5], which gives $\pi_2(x) \ll x(\log \log x)^2/(\log x)^2$ via sieve methods; this does not establish infinitude. The Hardy–Littlewood conjecture [6] predicts the asymptotic $\pi_2(x) \sim 2C_2 x/(\log x)^2$ where $C_2 \approx 0.6601$ is the twin prime constant, but this remains unproven. Recent unconditional work of Zhang [7] and Maynard [8] establishes infinitely many prime pairs with bounded gaps, but does not yield an explicit lower bound on $\pi_2(x)$. The bound $\pi_2(x) \gg \log x$ from Corollary 4.9 is very weak — far below the Hardy–Littlewood prediction — but it is a consequence of a purely additive conjecture about $A002822$, with no direct appeal to sieve methods or the distribution of primes in progressions.

Relation to other gap conjectures. Theorem 4.7 places BC in a broader context of conjectures asserting that gaps in a sequence grow no faster than the sequence itself.

Oppermann’s conjecture [13] asserts that for every integer $n > 1$ there is at least one prime in each of $(n(n-1), n^2)$ and $(n^2, n(n+1))$, implying that consecutive ordinary primes $p_k < p_{k+1}$ satisfy $p_{k+1} - p_k < \sqrt{p_k}$. BC makes an analogous assertion for twin prime witnesses: the gap $w_{n+1} - w_n < w_n$, so the next witness lies within a distance of the same order as the current value. The two conjectures are independent — Oppermann’s concerns the full prime sequence, BC concerns the twin prime witness sequence — but they share the same structural character: gaps are bounded by a function of order comparable to the current value. Computational verification of the weaker Legendre variant (a prime between n^2 and $(n+1)^2$) has been carried out to $n \leq 3.33 \times 10^{13}$ [14]. Oppermann’s conjecture itself remains open; claimed proofs circulating outside peer-reviewed literature are not generally accepted by the research community.

A density-scaling heuristic makes the $O(w_n)$ bound for twin prime witnesses plausible, and suggests it is not tighter than Oppermann implies. Ordinary primes near p occur with density $\sim 1/\ln p$; the Hardy–Littlewood conjecture gives twin prime pairs a density $\sim 2C_2/\ln^2 p$, a factor of $\sim \ln p$ thinner. In a Cramér-type probabilistic model, the expected maximum gap in a sequence with local density ρ scales as $\sim 1/\rho$. Replacing ρ by $\rho/\ln p$ therefore inflates the expected gap by $\ln p$, giving a heuristic maximum twin prime witness gap of $O(\sqrt{p} \ln p)$, comfortably within $O(w_n)$. Note that this argument also shows that an $O(\sqrt{w_n})$ bound — the direct twin prime analogue of Oppermann — would require a separate, stronger structural mechanism beyond density scaling alone.

Crucially, if Oppermann’s conjecture were known to imply $w_{n+1} - w_n < w_n$ directly — that is, if the ordinary prime gap bound could be transferred to the twin prime witness sequence — then the Twin Prime Conjecture would follow immediately, with no need for BC. The fact that no such implication is known is precisely why a dedicated conjecture like BC is needed: it asserts the $O(w_n)$ gap bound as an independent structural claim about the additive self-reinforcement of A002822, conditional on nothing beyond the infinite density of twin primes (TPC) itself. Put differently, BC is a plausible sharpening of TPC: if A002822 is infinite, BC says it grows without the gaps becoming catastrophically large.

Selcoe’s interval conjecture (SIC) [12] asserts that for every $k \geq 1$ there exists a twin prime witness $w \in A002822$ with $5k \leq w \leq 7k$. SIC implies the Twin Prime Conjecture directly and yields an $O(w_n)$ gap bound on consecutive witnesses, placing it in the same family as

BC. The two conjectures arrive at this bound by different mechanisms: BC via the additive sumset structure of A002822, SIC via a direct interval-covering assertion. Whether BC and SIC are equivalent or independent is an open question.

Heuristic and computational evidence for BC—including gap ratio plots, decomposition counts, stalling probability estimates, and an argument from the additive perspective—is developed in [1].

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